BYUI Coding Challenge for Adjunct Faculty Teaching DS 350

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Exploratory Data Analysis

```r
# Check directory
getwd()

# Importing libraries
library(tidyverse)
library(DataExplorer)
library(kableExtra)
```

You can use read_csv in Tidyverse, but technically vroom is a faster option.

```r
# Read in the data
baby <- read_csv("names_year_csv.csv")

# Faster way of reading in the data using vroom not tidyverse
baby <- vroom::vroom("names_year_csv.csv")

baby <- baby |>
  rename(Year = year, Name = name)
```

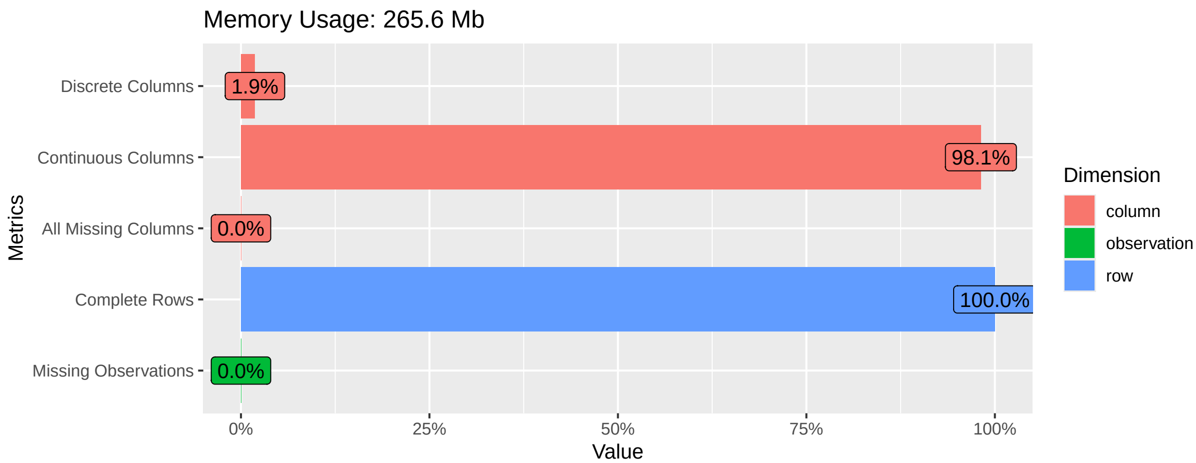
These are a few great ways to take a peek at the data.

```r
# Different options for quickly looking at the data
glimpse(baby)
head(baby)
```

```r
# Option for looking at the entire dataset in a separate tab
View(baby)
```

Using DataExplorer library we can see there are no missing observations, so we don't need to worry about that! There are also 33400 distinct baby names in this dataset.

```r
# Confirming there is no missing data
plot_intro(baby)
```




```r
# How many distinct names?
length(unique(baby$Name))
```

```
[1] 33400
```

Challenge questions

How does your name at your birth year compare to its use historically?

The answer for this question was pretty surprising for me. I wasn't expecting my birth year, 2002, to be the peak of of the name historically. The max was 2,679 babies whereas in 1947 only 5 babies were named Alexia!

The bar chart is a great way to visualize the data as you can clearly see the trend. It is interesting how there is a decrease after 2002, but a slight increase in 2013, then back down in the following years.

Funny enough, the only time I ever met someone else named Alexia was during my undergrad at BYU and we both happened to be statistics TAs :)

```
# Create a df for my name overtime
Alexia_Overtime <- baby |>
  arrange(desc(Total)) |>
  group_by(Name) |>
  filter(Name =="Alexia") |>
  select(Name, Year, Total)
```

```
#I want to take a look at the top 5 years
head(Alexia_Overtime, n=5) |>
  kbl(format = "latex") |>
  kable_classic_2(latex_options = "hold_position")
```

Name	Year	Total
Alexia	2002	2679
Alexia	2003	2484
Alexia	2004	2452
Alexia	2001	2284
Alexia	2005	2119

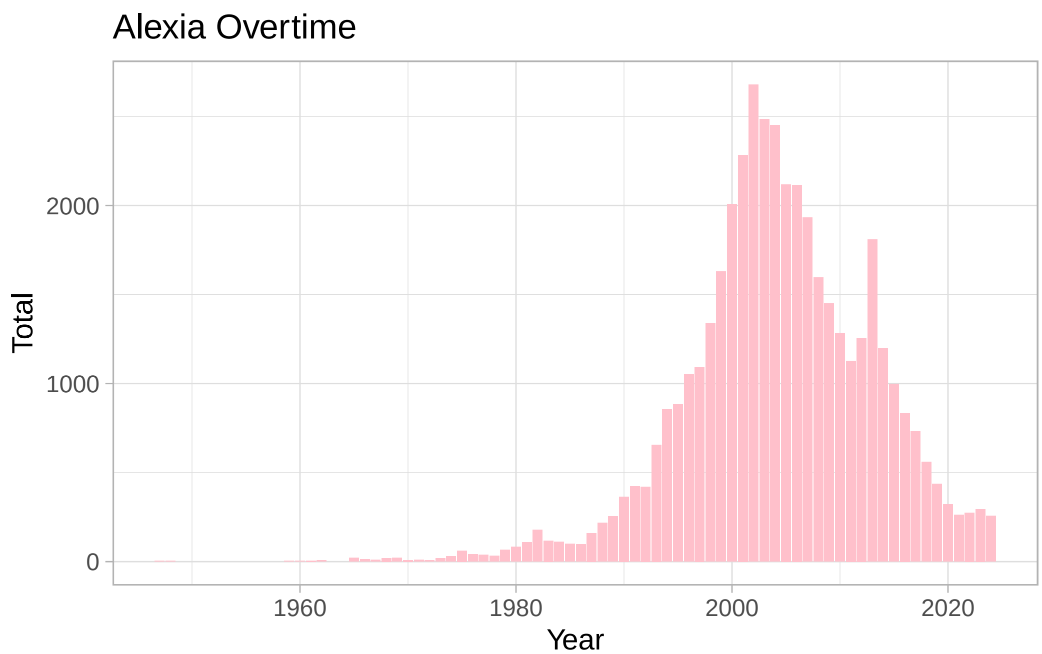
```
# Look at the bottom 5 years
tail(Alexia_Overtime, n=5)|>
  kbl(format = "latex") |>
  kable_classic_2(latex_options = "hold_position")
```

Name	Year	Total
Alexia	1961	6
Alexia	1947	5
Alexia	1948	5
Alexia	1959	5
Alexia	1960	5

```
# What my birth year looks like
Alexia_Overtime |>
  filter(Year == 2002) |>
  kbl(format = "latex") |>
  kable_classic_2(latex_options = "hold_position")
```

Name	Year	Total
Alexia	2002	2679

```
# How my name compares to it's use historically
ggplot(data = Alexia_Overtime, mapping = aes(x = Year, y=Total)) +
  geom_col(fill="pink") +
  labs(x = "Year", y="Total", title = "Alexia Overtime") +
  theme_light()
```



If you talked to someone named Brittany on the phone, what is your guess of their age? What ages would you not guess?

I would assume that Brittany was born in the late 80s-mid 90's. This would make her anywhere from 30-39 years old. I would not guess that she was in her 50's meaning she was born anytime from 1969-1975. Once we start looking past 1975, the popularity of the name increases with the peak in 1989 at 37,949 babies. This name is way more popular than mine has ever been! I am surprised we don't see an increase in the early 2000's. I would have guessed that Brittany Spears rising popularity would also make the name more trendy.

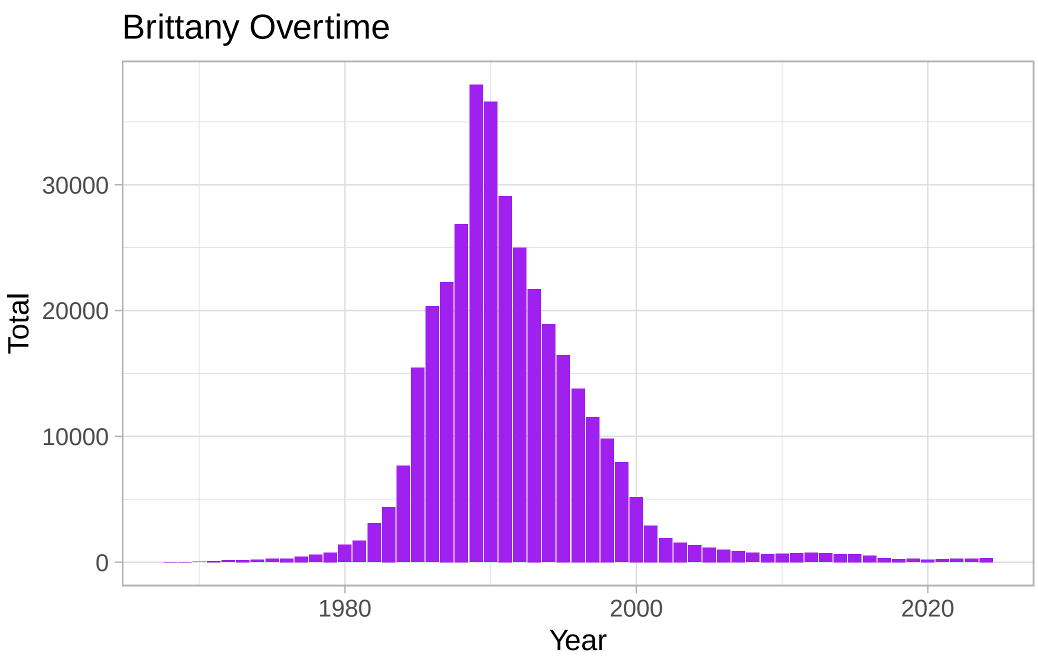
```
Brittany_Overtime <- baby |>
  arrange(desc(Total)) |>
  group_by(Name) |>
```

```r
  filter(Name =="Brittany") |>
  select(Name, Year, Total)

head(Brittany_Overtime) |>
  kbl(format = "latex") |>
  kable_classic_2(latex_options = "hold_position")
```

Name	Year	Total
Brittany	1989	37949
Brittany	1990	36600
Brittany	1991	29102
Brittany	1988	26867
Brittany	1992	24996
Brittany	1987	22268

```r
ggplot(data = Brittany_Overtime, mapping = aes(x = Year, y=Total)) +
  geom_col(fill="purple") +
  labs(x = "Year", y="Total", title = "Brittany Overtime") +
  theme_light()
```



Mary, Martha, Peter, and Paul are all Christian names. From 1920 - 2000, compare the name usage of each of the four names.

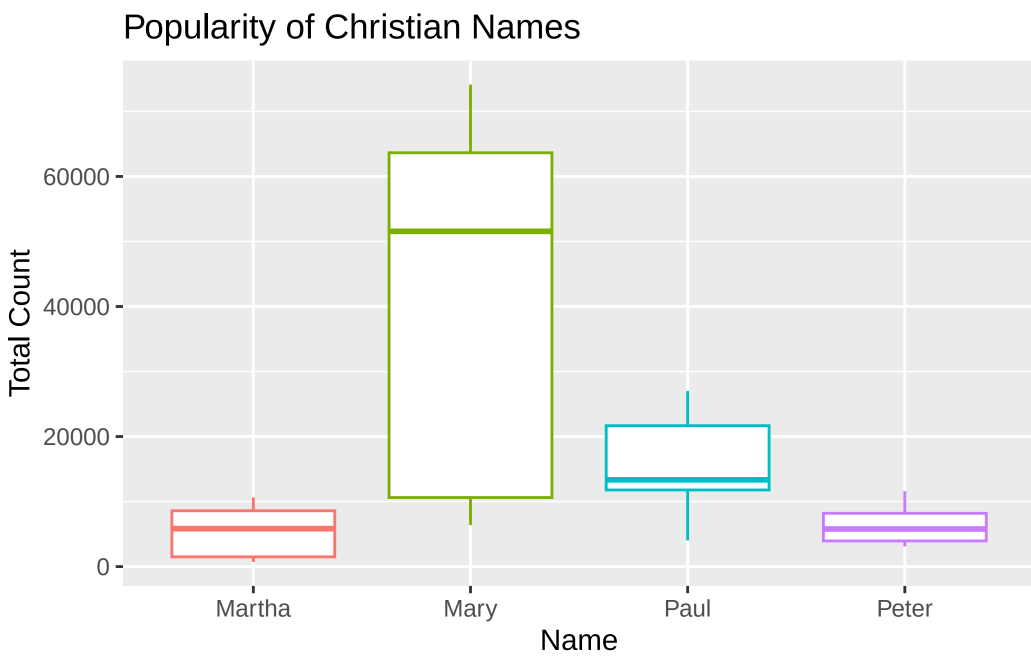
Below are several different visualizations to understand how these names compare to each other from the years 1920-2000s. As many would assume, Mary has been the favored name overall in comparison to Martha, Peter, and Paul. Overall, Mary is 7.5x more popular than Martha. Paul is preferred over Peter and the count of babies with the name Peter never has never surpassed Paul! Surprisingly, there are a couple years where Paul is actually higher than Mary but that doesn't start until about the mid 1970's.

```
# Make the table pretty using tibble?
Other_Overtime <- baby |>
  arrange(desc(Total)) |>
  group_by(Name) |>
  filter(Name =="Mary" | Name =="Martha" | Name =="Peter" | Name =="Paul",
         Year > 1920, Year < 2000) |>
  select(Name, Year, Total)

# Here is a visualization of all four names
ggplot(data = Other_Overtime, mapping = aes(x=Year, y=Total, color = Name)) +
  geom_line() +
  labs(x = "Year", y= "Total", color = "Name", title = "Christian Names Over the Years") +
  theme_bw()
```

```
# Paul is actually higher than Mary but that doesn't start until about the mid 1970's.
# Year 1967-1989
baby |>
  arrange(desc(Total)) |>
  group_by(Name) |>
  filter(Name =="Mary"| Name =="Paul",
         Year > 1920, Year < 2000) |>
  arrange(desc(Year)) |>
  select(Name, Year, Total)
```

```
ggplot(data = Other_Overtime, mapping = aes(x=Name, y=Total, color=Name)) +
  geom_boxplot() +
  labs(x = "Name", y= "Total Count", color = "Name", title = "Popularity of Christian Names")
  theme(legend.position = "none")
```




```
# All time count table
baby |>
  filter(
    Name %in% c("Mary", "Martha", "Peter", "Paul"),
    Year > 1920,
    Year < 2000
  ) |>
  group_by(Name) |>
```

```r
  summarise(across(-c(Year), sum)) |>
  select(Name, Total) |>
  arrange(desc(Total))  |>
  kbl(format = "latex") |>
  kable_classic_2(latex_options = "hold_position")
```

Name	Total
Mary	3131051
Paul	1226167
Peter	492348
Martha	415338

```r
# You could also write the filter line like this for the name
# name =="Mary" | name =="Martha" | name =="Peter" | name =="Paul",

# Overall, Mary is 7.5x more popular than Martha.
3131051/415338
```

```
[1] 7.538561
```

```r
# Christian Name totals by state
baby |>
  select(-Total) |>
  filter(
    Name %in% c("Mary", "Martha", "Peter", "Paul"),
    Year > 1920,
    Year < 2000
  ) |>
  group_by(Name) |>
  summarise(across(-c(Year), sum)) |>
  kbl(longtable = TRUE) |>
  kable_styling(latex_options = c("repeat_header", "hold_position"))
```

The tables below show the Christian name totals per state. This is a fun option to see which names have been historically the most popular for the state.

Name	AK	AL	AR	AZ	CA	CO	CT	DC	DE	FL
Martha	646	19205	9746	3301	26934	2940	2785	2120	644	10094
Mary	3517	96248	49272	20179	138843	26711	32230	20103	6956	59232

(continued)

Name	AK	AL	AR	AZ	CA	CO	CT	DC	DE	FL
Paul	1997	14247	10759	9280	95850	12637	21001	8229	3326	23354
Peter	1399	2013	721	3458	43322	3921	18773	3812	1173	8064

GA	HI	IA	ID	IL	IN	KS	KY	LA	MA	MD
23019	547	4445	750	15421	13468	4468	15305	7996	7750	4272
103026	4835	52955	8170	169636	81249	35720	84791	68001	89723	53700
17290	4465	18144	3705	65120	32417	11521	25475	19591	73381	19508
3176	2385	3809	806	29161	5561	1906	1514	5337	40490	5058

ME	MI	MN	MO	MS	MT	NC	ND	NE	NH	NJ
1568	10765	3311	10854	13122	759	20888	692	1535	1173	5297
12196	122865	63372	89030	73429	10433	111496	11075	25966	6827	72810
9292	55385	29557	28328	10067	3698	22949	5164	8472	6704	39339
6267	19205	14061	4727	1432	1369	3089	1867	1559	4630	32840

NM	NV	NY	OH	OK	OR	PA	RI	SC	SD	TN
3027	187	18275	21763	6544	1700	20453	1021	12807	675	19129
21395	2820	227899	166177	47085	17214	221284	12573	70221	12357	87775
6938	2046	122682	71557	13980	8873	90113	9570	11635	4622	21099
2049	718	105660	14958	1343	3811	30347	6057	1963	1069	1772

TX	UT	VA	VT	WA	WI
36441	908	12395	931	2550	3248
184241	11289	78200	6421	29622	76237
53757	8602	21002	3750	16417	31079
9797	1935	5065	2595	7474	17371

Think of a unique name from a famous movie. Plot that name's use over time and see how increases/decreases line up with the movie release.

For context, the first Hunger Games movie was released in 2012. By 2014 "The Hunger Games: Mockingjay – Part 1" was released with the series conclusion in 2015. We don't see the name appearing in the charts until 2014 which also happened to be the peak of the name, just 13 babies!

For fun I took a quick look at the name Han for the character Han Solo from Star Wars. The first star wars movie came out in 1977. Looks like this name wasn't used until 1979. The name reaches it's peak in 2018 with 17 babies. I would assume there is a correlation with "Solo: A Star Wars Story" being released that same year. Similar to Katniss, Han is still a pretty uncommon name.

```
# Picked two names just for fun
baby |>
  arrange(desc(Total)) |>
  group_by(Name) |>
  filter(Name =="Katniss") |>
  select(Name, Year, Total) |>
  kbl(format = "latex") |>
  kable_classic_2(latex_options = "hold_position")
```

Name	Year	Total
Katniss	2014	13
Katniss	2016	10
Katniss	2015	5

```
# As in Han Solo
head(baby |>
  arrange(desc(Year)) |>
  group_by(Name) |>
  filter(Name =="Han") |>
  select(Name, Year, Total) |>
  arrange(desc(Total))) |>
  kbl(format = "latex") |>
  kable_classic_2(latex_options = "hold_position")
```

Name	Year	Total
Han	2018	17
Han	2019	13
Han	2024	12
Han	2017	12
Han	2005	11
Han	1998	11

Make a private copy of this repo using the template button. Invite the GitHub user hathawayj to your repo. Use the Tidyverse to complete the challenge. Save your quarto file and other files created at time of rendering to your repository. Submit a link to your private repository.